

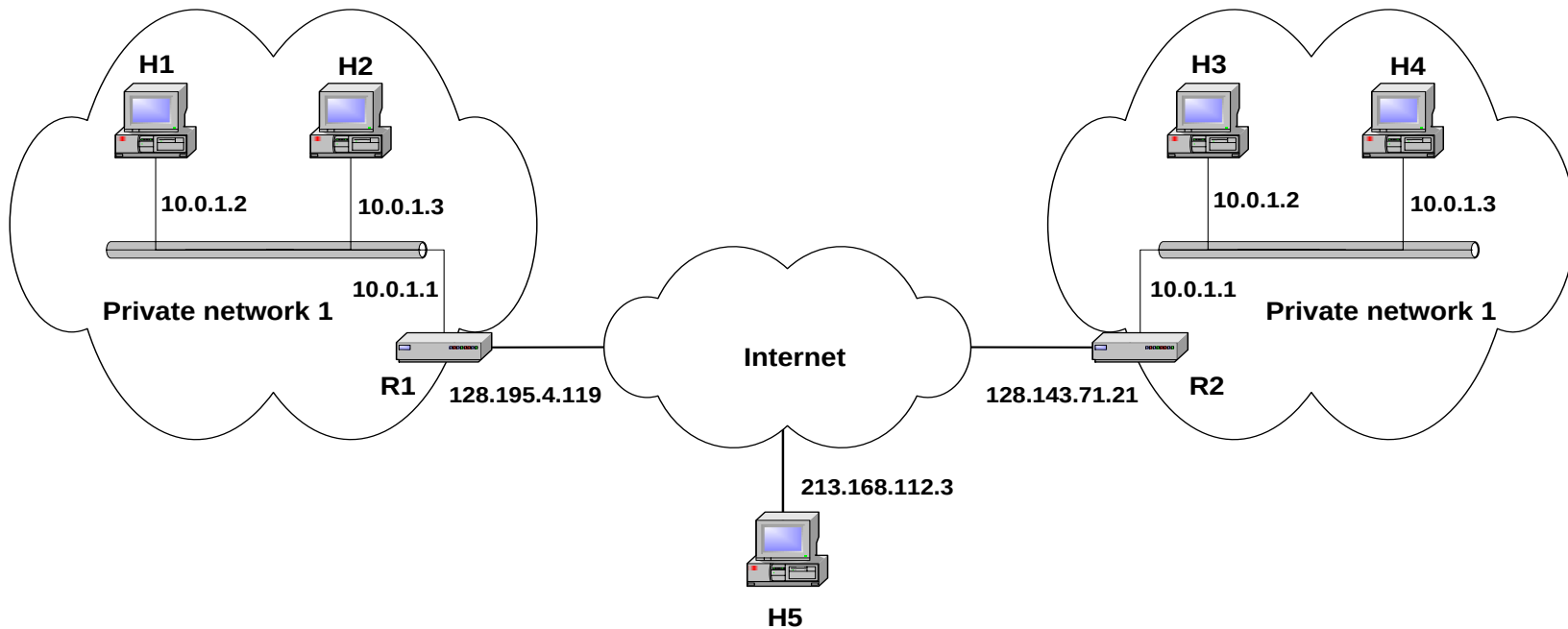
Network Address Translation (NAT)

Generally, private networks use addresses from the following experimental address ranges (*non-routable addresses*):

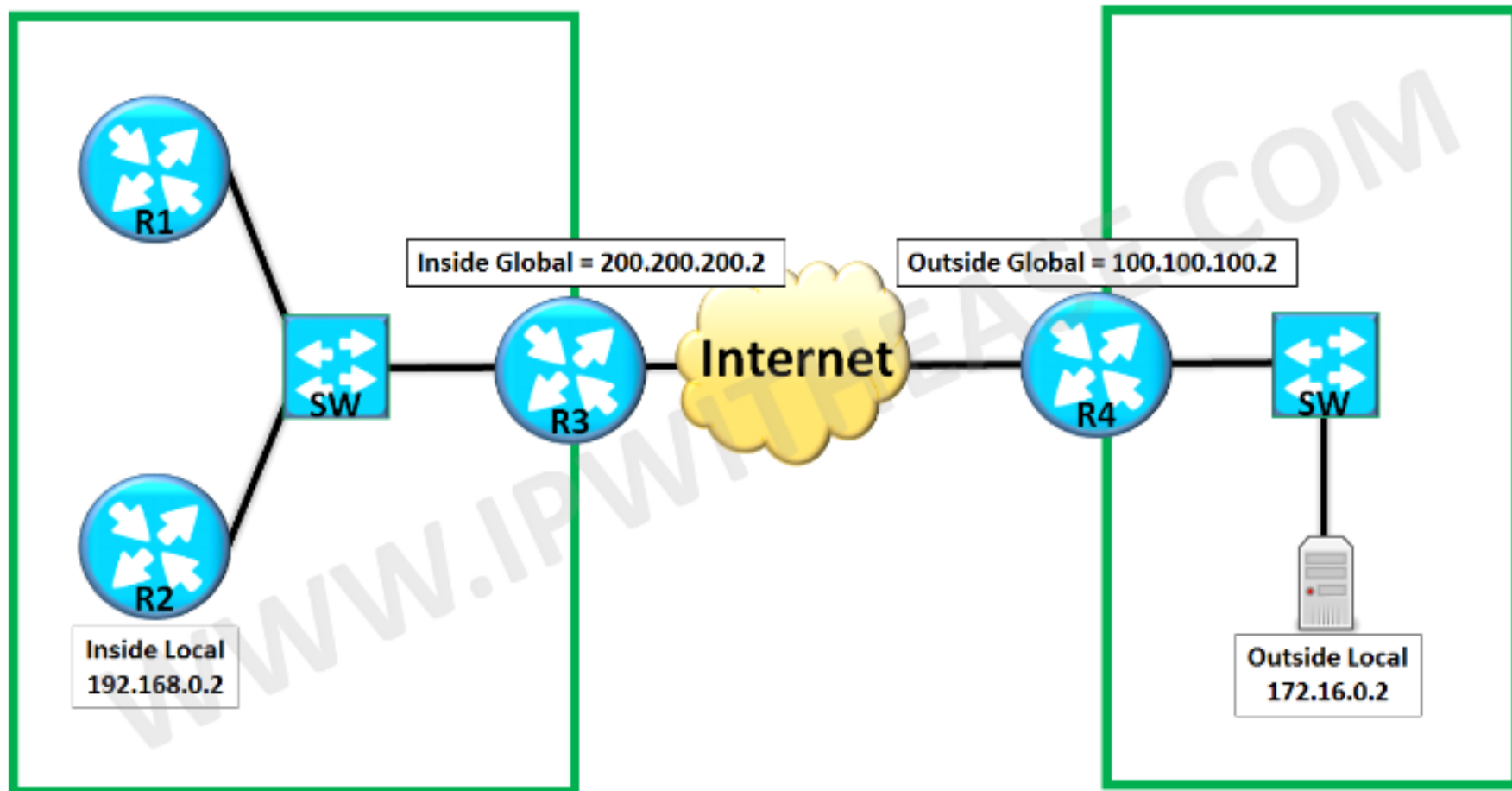
10.0.0.0 – 10.255.255.255

172.16.0.0 – 172.31.255.255

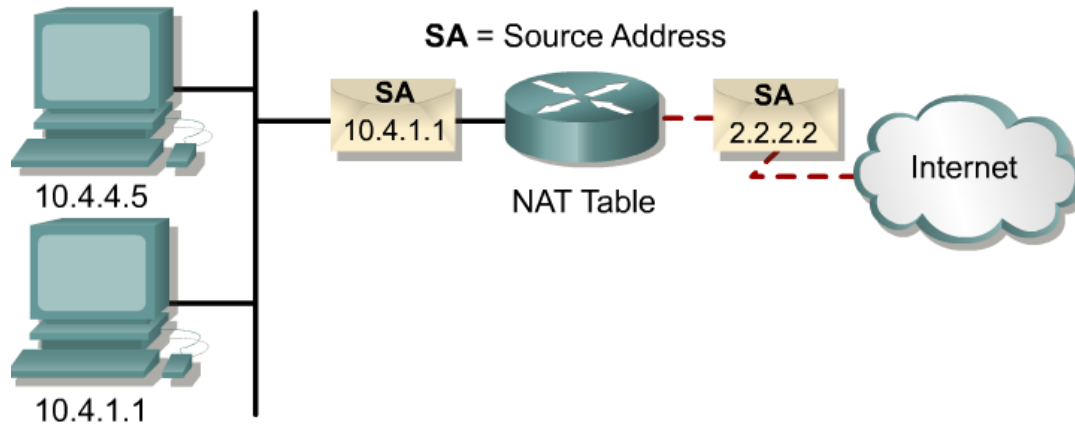
192.168.0.0 – 192.168.255.255



- A short term solution to the problem of the depletion of IP addresses
 - Long term solution is IPv6
 - CIDR (Classless InterDomain Routing) is a possible short term solution
 - NAT is another
- NAT is a way to conserve IP addresses
 - Can be used to hide a number of hosts behind a single IP address
 - Uses private addresses:
 - 10.0.0.0-10.255.255.255,
 - 172.16.0.0-172.32.255.255 or
 - 192.168.0.0-192.168.255.255



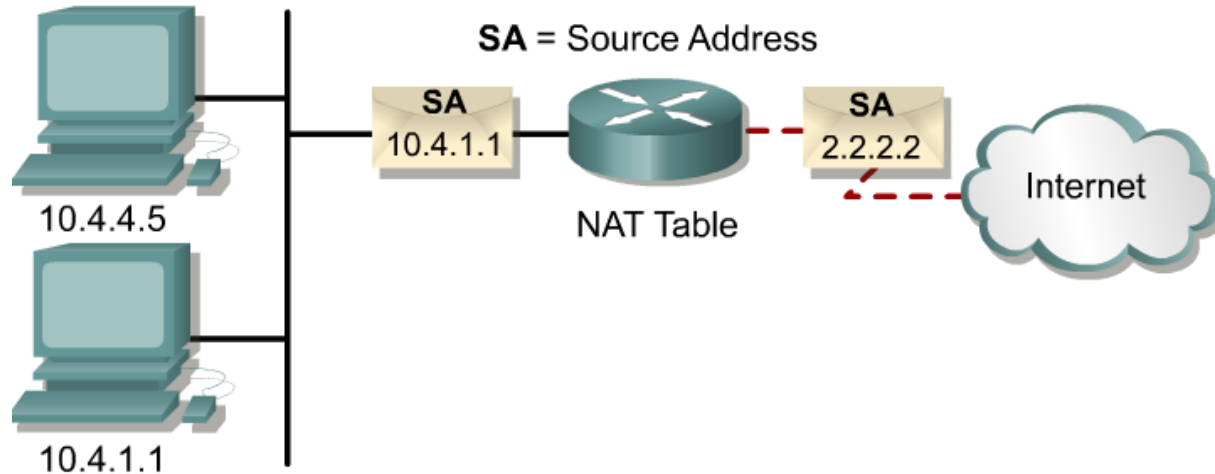
Static & Dynamic NAT



Inside Local IP Address	Global IP Address
10.4.4.5	2.2.2.3
10.4.1.1	2.2.2.2

- Static NAT is one-to-one address mapping.
- Dynamic NAT addresses are picked from a pool.

Masquerading (Network Address Port Translation)



Inside Local IP Address	Global IP Address
10.4.4.5	2.2.2.3 2.2.2.2 TCP Source Port 1923
10.4.1.1	2.2.2.2 2.2.2.2 TCP Source Port 1924

- PAT or address overloading
- m:1-translation
- The NAT router keeps track of the different conversations by mapping TCP and UDP port numbers.

Masquerading (Network Address Port Translation)

